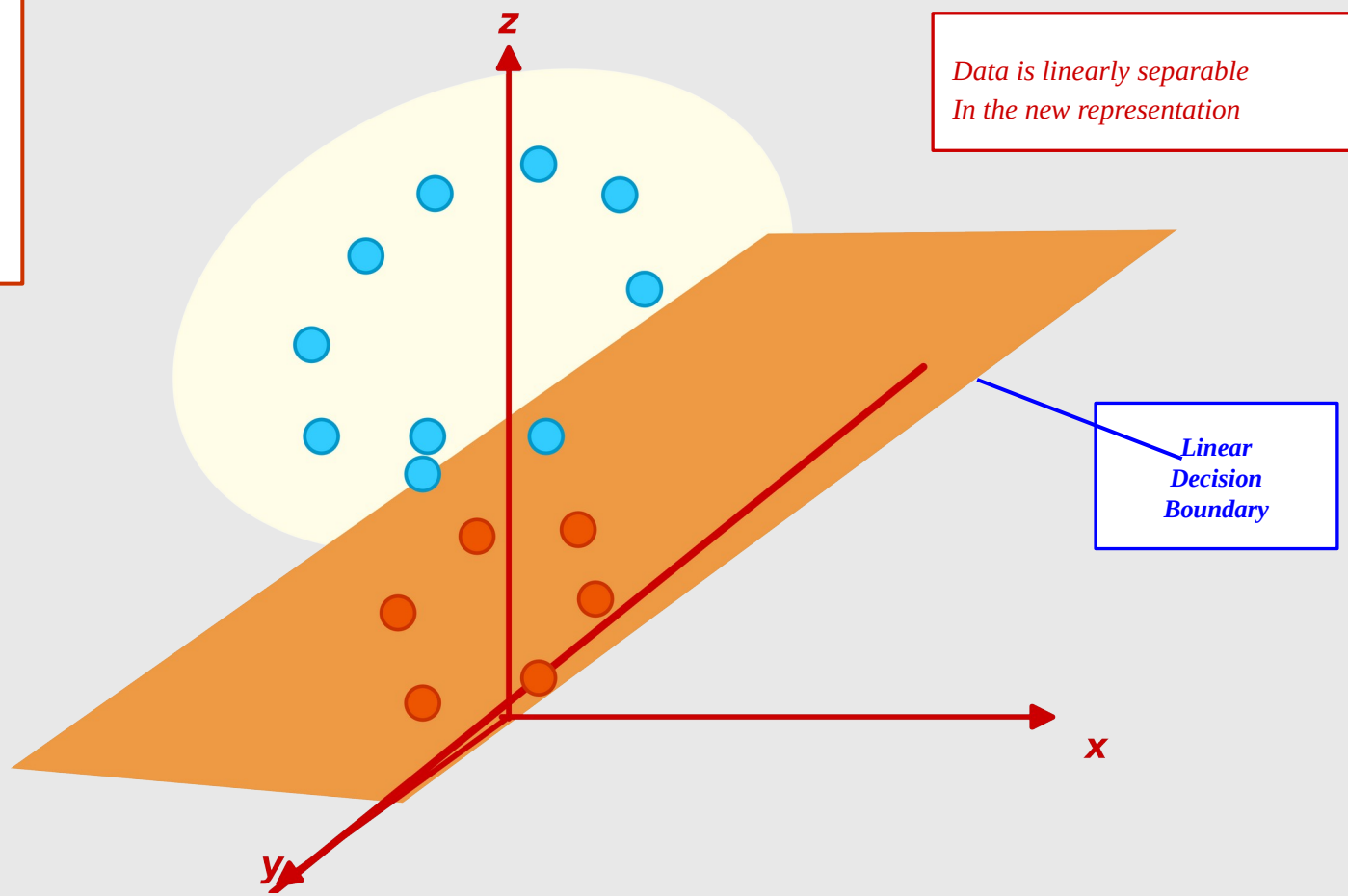
# Non-Linearly Separable Data

***Mapping***

$$\Phi(x, y) \rightarrow \begin{pmatrix} x^2 \\ \sqrt{2xy} \\ y^2 \end{pmatrix}$$




*Data is linearly separable*
*In the new representation*

***Linear Decision Boundary***